Dung beetle biogeography

100
90
80
70
60
50
40
30
20
10
0
Helictopleurus fasciolatus suturalis STL10234
Helictopleurus sp STL10248
Helictopleurus cf. Furcicornis STL10332
Helictopleurus furcicornis STL10169
Helictopleurus fasciolatus fasciolatus STL10324
Helictopleurus fasciolatus pseudofasciolatus STL10333
Helictopleurus fasciolatus obscurus STL10334
Helictopleurus neuter STL10327
Helictopleurus fissicollis morettoi STL10237
Helictopleurus fissicollis australis STL10246
Helictopleurus fissicollis STL28
Helictopleurus fulgens STL10168
Helictopleurus nigricans STL10238
Helictopleurus fulgens purpuricollis STL10239
Helictopleurus hanskii STL10247
Helictopleurus viridans viridans STL10235
Helictopleurus viridans pseudoviridans STL10236
Helictopleurus politicollis STL5029
Helictopleurus splendidicollis STL5020
Helictopleurus unifasciatus STL5030
Helictopleurus quadripunctatus STL5027
Helictopleurus perrieri STL10329
Helictopleurus rudicollis STL10173
Helictopleurus undatus STL10331
Helictopleurus cribricollis STL10243
Helictopleurus giganteus STL10177
Helictopleurus sinuatocornis STL10330
Helictopleurus aurocupreus STL5025
Helictopleurus coruscus STL10209
Helictopleurus marsyas STL5026
Helictopleurus nicollei STL10319
Helictopleurus sp group semivirens STL10214
Helictopleurus dorbignyi STL10240
Helictopleurus steineri STL10264
Helictopleurus cf steineri n sp STL10211
Helictopleurus semivirens STL10242
Helictopleurus cambeforti STL10250
Helictopleurus carbonarius STL10249
Helictopleurus peyrierasi STL5021
Helictopleurus viettei STL10241
Helictopleurus sicardi ST013
Helictopleurus neoamplicollis STL5024
Helictopleurus vadoni STL10244
Cyptochirus sp1 STL10112
Scaptocnemis segregis STL10251
Oniticellus planatus STL10119
Liatongus sp STL10290
Oniticellus cinctus ST004
Euoniticellus fulvus STL10132
Macronthophagus diabolicus STL10291
Onthophagus sl laevis STL10302
Onthophagus Sunenaga streltsovi cf STL10303
Onthophagus Serrophorus seniculus STL10110
Onthophagus cf muelleri STL10298
Parascatonomus anceyi STL10091
Onthophagus Parascatonomus tricornis STL10111
Proagoderus cf lanista STL10079
Proagoderus sapphirinus STL10098
Onthophagus pipitzi STL5018
Onthophagus Paraphanaeomorphus aff phanaeiformis STL10113
Onthophagus Hikidaeus waterstradti STL10308
Onthophagus sl batillifer STL10301
Onthophagus Micronthophagus sp1 STL10115
Onthophagus Onthophagiellus cf sp1 STL10304
Onthophagus sl vulpes STL10307
Onthophagus Gibbonthophagus rectecornutus STL10114
Onthophagus sstr obscurior STL10305
Onthophagus taurus ST002
Onthophagus sl baeri STL10084
Onthophagus sl pilosus STL10101
Onthophagus sl giraffa STL10130
Eusaproecius sp1 STL10256
Onthophagus cf.bicavifrons STL10006
Onthophagus sl cf quadrimaculatus STL10046
Onthophagus sl leucopygus STL10100
Caccobius schereberi ST001
Cleptocaccobius viridicollis STL10099
Onthophagus Bicornonthophagus interstitialis STL10081
Onthophagus Euonthophagus flavimargo STL10109
Onthophagus Pseudophanaeomorphus quasijohkii aff STL10299
Onthophagus sl borneensis STL10306
Onthophagus sl vinctus STL10080
Onthophagus Furconthophagus variegatus STL10083
Onthophagus sl lamelliger STL10131
Caccobius binodulus STL10300
Onthophagus sl sp1 cf gr STL10315
Onthophagus Palaeonthophagus gibbulus STL10133
Amietina larrochei STL10208
Phalops flavocinctus STL10045
Digitonthophagus gazella STL10213
Onthophagus probus STL10015
Hamonthophagus depressus STL10013
Kurtops signatus STL10043
Hammondantus psammophilus STL10321
Pycnopanelus krikkeni STL10076
Onitis alexis STL10041
Anonychonitis freyi STL10095
Megalonitis bohemani STL10318
Onitis caffer STL10042
Heteronitis ragazzii STL10032
Onitis uncinatus STL10031
Cheironitis scabrosus STL10096
Macroderes foveatus STL10040
Macroderes sp1 STL10328
Xinidium dentilabris STL10004
Xinidium dewitzi STL10008
Dwesasilvasedis medinae STL10265
Catharsius sp. STL10033
Copridaspidus peregrinus STL10258
Catharsius cf molossus STL10120
Metacatharsius opacus STL10017
Metacatharsius sp1 STL10007
Epilissus cuprarius STL5011
Epilissus sp morpho 1 STL26
Epilissus aff morio group STL10184
Epilissus morio STL5007
Epilissus fuscovirens STL5010
Epilissus alluaudi STL5009
Epilissus aff planatus STL10187
Epilissus apotolamproides STL5008
Epilissus aff antoetrae STL10180
Epilissus sp1 STL10186
Epilissus sp morpho 2 STL48
Epilissus splendidus splendidus ST008
Epilissus splendidus cf delphinensis STL10185
Epilissus aff semilissus STL10181
Arachnodes sp7 cf insularis STL10201
Arachnodes sp4 STL5016
Arachnodes pusillus STL5015
Arachnodes sp. morpho 1 STL45
Arachnodes sp STL10216
Arachnodes aff saprinoides STL10195
Arachnodes saprinoides STL10151
Arachnodes sp2 STL5013
Arachnodes aff robinsoni STL10194
Arachnodes sp5 STL10217
Arachnodes sp1 STL5012
Arachnodes sp3 STL10218
Arachnodes sp morpho 2 STL37
Arachnodes sp6 STL10200
Arachnodes oberthuri STL10144
Apterepilissus ovalis STL10325
Copris (Paracopris) sp. STL10022
Copris fidius ST005
Copris gilleti STL10028
Copris nevinsoni STL10021
Copris sp2 STL10025
Copris urus STL10024
Litocopris muticus STL10027
Copris sl diversus STL10245
Microcopris sp1 STL10023
Pedaria sp1 STL10016
Pedaria sp STL10323
Pedaria sp2 STL10019
Onycothechus tricornigeris STL10225
Coptodactyla brooksi CopbroUCE
Thyregis relictus Thyrel UCE
Amphistomus primonactus Amppri UCE
Diorygopyx dudplodentatus Diodud UCE
Temnoplectron lewisensis Temlew UCE
Heliocopris sp STL10271
Heliocopris sp STL10292
Synapsis sp STL10129
Nanos morpho 2 STL35
Nanos aff mirjae STL10189
Nanos dubitatus STL5001
Nanos aff ST009
Apotolamprus sp STL30
Nanos aff viridissimus STL10191
Nanos sp8 STL10204
Nanos sp10 STL10206
Nanos sp7 STL5005
Nanos sp11 STL10207
Nanos sp9 STL5004
Nanos sp STL39
Nanos sp6 STL5002
Nanos aff rubrosignatus STL10142
Nanos vadoni ST011
Apotolamprus vadoni STL10135
Apotolamprus quadrimaculatus STL10139
Apotolamprus aff ambohitsitondronensi STL10140
Apotolamprus quadrinotatus STL5006
Apotolamprus aff latipennis STL10197
Apotolamprus aff sparsepunctatus STL10198
Apotolamprus aff metallicus STL10160
Apotolamprus sp1 STL10146
Apotolamprus perinetensis STL10161
Apotolamprus sp. morpho 1 STL31
Arachnodes aff metallicus STL10196
Nanos sp2 STL10153
Nanos sp4 STL10215
Nanos sp5 STL10202
Nanos aff humeralis STL10190
Cambefortantus pauliani STL32
Nanos bicoloratus STL10155
Nanos aff Bicoloratus STL10137
Nanos sp1 STL10147
Nanos aff Pseudominutus STL10156
Scarabaeolus intricatus STL10044
Scarabaeus westwoodi STL10034
Sceliages hippias STL10030
Drepanopodus proximus STL10001
Pachylomera femoralis STL10128
Scarabaeus n sp1 STL10210
Pachysoma hippocrates STL10127
Kheper nigroaeneus STL10036
Circellium bacchus STL10270
Chalconotus convexus STL10294
Gyronotus pumilus STL10003
Garreta unicolor STL10077
Paragymnopleurus sp STL10092
Gymnopleurus aenescens STL10097
Paragymnopleurus sp STL10293
Nesovinsionia vinsoni STL4
Tanzanolus leleupi STL10223
Janssensantus cf inexpectatus STL10103
Janssensantus mabuensis n sp STL10102
Panelus sp3 STL51
Panelus rhodesiensis STL10037
Sisyphus (Neosisyphus) infuscatus STL14
Sisyphus (Neosisyphus) spinipes STL17
Sisyphus (Neosisyphus) rubrus STL12
Sisyphus (Neosisyphus) barbarossa STL8
Sisyphus (Neosisyphus) calcaratus STL7
Sisyphus (Neosisyphus) mirabilis STL11
Sisyphus umbraphilus STL10
Sisyphus sp10 STL42
Sisyphus sp6 STL23
Sisyphus sp8 STL40
Sisyphus schaefferi STL44
Sisyphus australis STL13
Sisyphus sordidus STL6
Sisyphus inconspicuus STL16
Sisyphus gazanus STL9
Sisyphus manni STL15
Sisyphus muricatus STL5
Sisyphus sp1 STL18
Nesosisyphus regnardi STL2
Nesosisyphus rotundatus STL52
Nesosisyphus vicinus STL1
Nesosisyphus pygmaeus STL3
Epirinus sp. 1 STL43
Epirinus flagellatus STL10118
Epirinus cf.silvestris STL10018
Bohepilissus sp1 STL10279
Epactoides femoralis STL10145
Epactoides aff major STL10163
Epactoides major STL47
Epactoides sp STL5017
Epactoides jounii STL33
Epactoides hanski STL29
Epactoides sp2 STL10157
Epactoides aff Lissus STL10134
Epactoides semiaeneus STL10136
Epactoides aff Semiaeneus STL10158
Epactoides olsoufieffi STL10141
Epactoides mangabeensis STL10154
Epactoides sp3 STL10162
Grebennikovius lupanganus sp n1 CNCCOLVG00004076
Ochichanton sp1 STL25
Odontoloma planatum STL10048
Odontoloma sp1 STL10288
Endroedyolus paradoxus STL10049
Byrrhidium namaquensis STL10268
Dicranocara deschodti STL10269
Haroldius sp1 STL10266
Haroldius sp2 STL10273
Sarrophorus costatus ST006
Coptorhina klugii STL10039
Frankenbergerius armatus STL10322
Paraphytus cf hindu STL10093
100
90
80
70
60
50
40
30
20
10
0
Age (Ma)
A Africa
E Eurasia-Oriental
M Madagascar
U Australia
S Paleoislands
R Mauritius
A.E
A.M
A.U
A.S
A.R
E.M
E.U
E.S
E.R
S.R